

The Actualization Theory

A Reconstruction of Physics from
Difference, Actualization and Spectrum

Prediction Registry

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The Actualization Theory (AT) — Prediction Registry v1.0

Companion document to The Actualization Theory (AT) v2.0 (Canonical)

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Abstract

This document is the official *Prediction Registry* of The Actualization Theory (AT): a standalone, publication-grade registry of the quantitative and qualitative predictions derived from the canonical theory, compiled before future experimental confirmation. Every entry is derived from the accepted canonical derivations of the monograph (Difference $\rightarrow$ Actualization $\rightarrow$ Inevitable Spectrum $\rightarrow$ Physics), carries its derivation source, its dependency chain, its numerical value (where applicable), its uncertainty, its validation status, and its falsification criterion. The registry uses no new physics, no new fits, and no retrospective reinterpretation; it records the canonical content as published. Statuses distinguish CONFIRMED (pre-registered target verified), CONSISTENT (value matches current observation), PENDING (awaiting validation), and FALSIFIED (contradicted by observation). No registered prediction is currently falsified.

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Part I

Introduction

1 Purpose of the Prediction Registry

The Prediction Registry exists to fix, in advance of experiment, the falsifiable content of The Actualization Theory. Its purposes are:

- to record every quantitative and qualitative prediction derived from the canonical theory, with its derivation source and dependency chain;
- to fix the numerical values and uncertainty ranges before future confirmation, so that confirmation or contradiction is genuine independent evidence and not post-hoc fitting;
- to document, per prediction, the experimental test, the current evidence, and the falsification criterion;
- to serve as the immutable, version-controlled reference for external validation (the theory’s pre-registration program, QG190–QG193, and its outcome dashboard, QG202);
- to be suitable for independent Zenodo publication and future version-controlled updates.

The registry is *assembly only*: it contains no new physics, no new derivations, no new fits, and no speculation. Every entry is drawn from the accepted canonical results of the monograph V2.0 and the claim-classification registry.

2 Validation Philosophy

The Actualization Theory distinguishes *derivation* from *validation* (monograph Ch. 14, Theorem c14:thm:derived-vs-validated): a quantity may be derived without being validated, and its derivation is complete before its validation is possible. This registry records derived predictions and their validation status on that basis.

The statuses used throughout are:

- CONFIRMED: a pre-registered target verified by observation (the prediction was frozen before the data).
- CONSISTENT: the derived value matches current observation within the stated deviation (including calibrated correspondences and documented fits). These are retrospective consistency records, not pre-registered confirmations.
- PENDING: awaiting validation — the current experimental frontier.
- FALSIFIED: contradicted by observation. As of this version, no registered prediction is falsified.

The classification of each prediction’s *derivational* status — theorem, necessity, correspondence, calibration, hosted, fit — is preserved from the claim-classification registry and disclosed in each record. In particular, entries classified as *calibration* carry an anchor (v , m_e , or SI conversion) and entries classified as *fit* are labelled as fitted readouts, not derivations.

3 Pre-Registration Principles

The theory maintains an immutable prediction registry (QG193) containing three pre-registered predictions with frozen values and explicit falsification conditions: P1 (a 106 GeV resonance), P2 (the $0\nu\beta\beta$ effective Majorana mass), and P3 (the sector-ladder spectrum). The pre-registration principles are:

- **Frozen values:** each pre-registered prediction’s value is fixed before future data; no future phase may modify it (registry lock, QG193).
- **Forbidden inputs:** pre-registered predictions assert that no experimental limit, detector sensitivity, fitted mass, or future measurement entered the derivation (forbidden-input guards, QG190–QG192).
- **State advance only:** the outcome state may advance forward (PENDING $\rightarrow$ SUPPORTED $\rightarrow$ CONFIRMED, or PENDING $\rightarrow$ DISFAVORED $\rightarrow$ FALSIFIED); frozen values never change.
- **Temporal independence:** confirmation of a pre-registered prediction is genuine independent evidence because the value was frozen before the observation.

The three pre-registered predictions are registered in full in Section 20. All other entries in this registry are derived values with documented consistency or pending status.

Part II

Standard Model Predictions

4 Gauge Sector

AT-P001

Status: CONSISTENT

Title: Gauge structure $1 + 3 + 8$ (dimensional correspondence).

Derived From: Difference $\rightarrow$ Actualization $\rightarrow$ Spectrum (D96) $\rightarrow$ degree-12 connectivity of $C_{96}(\pm 1, \dots, \pm 6)$.

Prediction: The gauge sector of the standard model, $U(1) \times SU(2) \times SU(3)$, corresponds dimensionally to the D96 sector structure: one $U(1)$, three $SU(2)$, and eight $SU(3)$ generators, $1 + 3 + 8 = 12$, the degree of the C_{96} generator ring. The gauge groups and their Lie algebras are *hosted*, not derived (claim classification: dimensional correspondence).

Experimental Test: Count of gauge generators in the observed standard model.

Current Evidence: The standard model carries exactly 12 gauge generators (1 photon + 3 weak + 8 gluons). The correspondence is dimensional; the groups are hosted.

Future Validation Path: No experimental change expected; the claim is a dimensional correspondence, not a prediction of new group content.

Falsification Criterion: Observation of a gauge sector not matching the 1 + 3 + 8 dimensional count (not currently conceivable at low energy).

AT-P002**Status: CONSISTENT**

Title: Gauge couplings as spectral ratios.

Derived From: Difference $\rightarrow$ Actualization $\rightarrow$ Spectrum (D96) $\rightarrow$ spectral constants Σm , $\Sigma\sqrt{m}$, $\#d$; couplings $1/\alpha_{\text{em}} = \Sigma m + \#d = 137$ (fitted readout), $\alpha_{\text{weak}} = 3/\Sigma m$, $\alpha_{\text{strong}} = 8/\Sigma\sqrt{m}$, $\sin^2 \theta_W = 0.2316$.

Prediction: The weak and strong couplings correspond to D96 spectral ratios (3/95, 8/64.08); the fine-structure inverse is the fitted readout $137 = 95 + 42$. None of the couplings carries a defined renormalization scale (claim classification: $1/\alpha_{\text{em}} = \text{fit}$; α_{weak} , $\alpha_{\text{strong}} = \text{correspondence}$).

Experimental Test: Measurement of the gauge couplings and the Weinberg angle.

Current Evidence: $\sin^2 \theta_W = 0.2316$ matches the measured value; the weak and strong ratios match the observed couplings at the natural scale; $1/\alpha_{\text{em}} = 137$ matches the low-energy fine-structure constant (documented fit).

Future Validation Path: Precision running of α_s to the Z pole: spectral $\alpha_s = 8/\Sigma\sqrt{m} = 0.1248$ vs. PDG $\alpha_s(M_Z) = 0.1184$ (dev. 5.4%); improvement of the measured low-energy α_{em} .

Falsification Criterion: A measured gauge coupling inconsistent with the spectral ratios at the natural scale beyond the stated correspondence accuracy.

AT-P003**Status: CONSISTENT**

Title: Weak boson masses from the weak scale.

Derived From: Difference $\rightarrow$ Actualization $\rightarrow$ Spectrum (D96) $\rightarrow$ weak scale $v = (\Sigma m + \#d) \ln(\text{span}) = 254$ GeV.

Prediction: $M_W = 80.1$ GeV and $M_Z = 91.4$ GeV from the Weinberg projection, with $\rho = M_W^2/(M_Z^2 \cos^2 \theta_W) = 1$ exactly at tree level.

Experimental Test: Measurement of the W and Z masses (LEP/ATLAS/CMS).

Current Evidence: Measured $M_W \simeq 80.4$ GeV, $M_Z = 91.19$ GeV; the reconstruction agrees within the calibration accuracy (claim classification: calibration via the anchor v).

Future Validation Path: Future W -mass refinements at HL-LHC; M_Z is precisely known.

Falsification Criterion: A measured M_W/M_Z ratio inconsistent with $\rho = 1$ beyond the calibration accuracy.

AT-P004**Status: CONSISTENT**

Title: Precision electroweak observables.

Derived From: Difference $\rightarrow$ Actualization $\rightarrow$ Spectrum (D96) $\rightarrow$ spectral constants; $\sin^2 \theta_{\text{eff}} = \#g/(2\Sigma m) = 0.23158$; $\Gamma_Z = M_H \cos \theta_W / \#g = 2.4953$; $\Gamma_W = \sigma_{\text{occ}}^2 / (\text{occMom} \cdot \lambda_2) = 2.0852$;

$\Gamma_H = \lambda_2/\Sigma m = 4.07 \text{ MeV}$; $R_b = \text{span} \cdot g_2 \cdot \sin^4 \theta_W = 0.2163$; $A_{FB}^b = (\lambda_H/\lambda_2)^2 = 0.0992$; $A_{FB}^\ell = M_H/(M_W M_Z) = 0.0171$.

Prediction: The precision electroweak observables (effective Weinberg angle, Z and W widths, Higgs width, R_b , forward-backward asymmetries) are read as spectral ratios (claim classification: correspondence).

Experimental Test: Measurements of the Z -pole observables, R_b , and the asymmetries at LEP/SLC and LHC.

Current Evidence: $\sin^2 \theta_{\text{eff}} = 0.23158$ matches the measured 0.23153 within 0.03%; the widths and R_b match within the stated deviations (0.004%–0.05%).

Future Validation Path: HL-LHC precision runs; any improvement of the R_b and asymmetry measurements.

Falsification Criterion: A measured precision observable deviating from its spectral ratio beyond the stated accuracy.

5 CKM and PMNS

AT-P005

Status: CONSISTENT

Title: CKM matrix as octave-transition read.

Derived From: Difference $\rightarrow$ Actualization $\rightarrow$ Spectrum (D96) $\rightarrow$ octave transitions: $V_{us} = \#d/(2\Sigma m)$, $V_{cb} = (\Omega_0/\Omega_2)^{\delta_d}$, $V_{ub} = 2V_{cb} \cdot (\text{occ}_0/\text{occ}_2)$; $\Omega_0 = 0.924$, $\Omega_2 = 3.385$ are the octave-family centers (not the Laplacian modes $\omega_k = \sqrt{\lambda_k}$), $\delta_d = 2.449$ the down-sector effective dimension.

Prediction: The quark mixing matrix elements are spectral correspondences matched to the measured CKM matrix with mean deviation 0.58% (claim classification: correspondence).

Experimental Test: Measurement of the CKM elements in weak decays.

Current Evidence: The three reconstructed elements match the measured values with mean deviation 0.58%.

Future Validation Path: Improvement of V_{cb} and V_{ub} measurements (Belle II, LHCb); the ratios are fixed spectral forms.

Falsification Criterion: A measured CKM element deviating from its spectral ratio beyond the stated correspondence accuracy.

AT-P006

Status: CONSISTENT

Title: CKM CP violation and the Jarlskog invariant.

Derived From: Difference $\rightarrow$ Actualization $\rightarrow$ Spectrum (D96) $\rightarrow$ chiral circulation: $\sin \delta = \text{occ}_{\text{top}}/\Sigma m = 0.9158$, $\delta_{CP} = 66.3^\circ$, $J = 3.139 \times 10^{-5}$.

Prediction: The CKM CP phase and the Jarlskog invariant are spectral-ratio reads (claim classification: correspondence).

Experimental Test: Measurement of CP-violating observables in the quark sector.

Current Evidence: $\delta_{CP} \simeq 66^\circ$ is consistent with the global CKM fit; $J = 3.139 \times 10^{-5}$ matches the measured $J \simeq 3 \times 10^{-5}$.

Future Validation Path: Precision CP measurements at Belle II and LHCb.

Falsification Criterion: A measured J inconsistent with the spectral value beyond the stated accuracy.

AT-P007

Status: CONSISTENT

Title: PMNS matrix as T3-only read.

Derived From: Difference $\rightarrow$ Actualization $\rightarrow$ Spectrum (D96) $\rightarrow$ T3-only access reads: $\theta_{12} = 33.35^\circ$, $\theta_{23} = 49.72^\circ$, $\theta_{13} = 8.34^\circ$, $\delta_\nu = 66.4^\circ$, mean deviation 1.5%.

Prediction: The lepton mixing angles and the Dirac CP phase are spectral correspondences (claim classification: correspondence).

Experimental Test: Measurement of the PMNS angles in neutrino oscillations.

Current Evidence: The angles match the measured values with mean deviation 1.5%.

Future Validation Path: Improved θ_{23} and δ_ν determination (DUNE, JUNO, Hyper-K); the $\delta_\nu = 66.4^\circ$ read is a continuous spectral-ratio phase, not lattice-restricted.

Falsification Criterion: A measured PMNS angle deviating from its spectral read beyond the stated correspondence accuracy.

6 Neutrino Sector

AT-P008

Status: CONSISTENT

Title: Neutrino mass splittings.

Derived From: Difference $\rightarrow$ Actualization $\rightarrow$ Spectrum (D96) $\rightarrow$ closed-form laws: $\Delta m_{21}^2 = (1/\Sigma\sqrt{m})^2/(\text{span}/2) = 7.607 \times 10^{-5} \text{ eV}^2$ (dev. 1.02%), $\Delta m_{31}^2 = \sin^2 \theta_W / \Sigma m = 2.44 \times 10^{-3} \text{ eV}^2$ (dev. 0.71%).

Prediction: The solar and atmospheric splittings are closed-form D96 ratios, calibrated to eV^2 by an implicit scale (claim classification: correspondence + calibration).

Experimental Test: Measurement of the splittings in solar, atmospheric, and reactor neutrino oscillations.

Current Evidence: $\Delta m_{21}^2 = 7.607 \times 10^{-5}$ and $\Delta m_{31}^2 = 2.44 \times 10^{-3} \text{ eV}^2$ match the measured values within the stated deviations.

Future Validation Path: Precision oscillation experiments (JUNO, DUNE, Hyper-K).

Falsification Criterion: A measured splitting deviating from its closed-form ratio beyond the stated accuracy.

AT-P009

Status: PENDING

Title: Absolute neutrino masses (normal ordering).

Derived From: Difference $\rightarrow$ Actualization $\rightarrow$ Spectrum (D96) $\rightarrow$ absolute mass law: $m_1 = 0$ (zero-mode, normal ordering), $m_2 = 1/(\Sigma\sqrt{m} \cdot \sqrt{\text{span}/2}) = 8.7216 \text{ meV}$ (phys. 8.72), $m_3 = \sqrt{\#g}/(\Sigma m\sqrt{2}) = 49.3728 \text{ meV}$ (phys. 49.4); $\Sigma m_\nu = 0.0581 \text{ eV}$.

Prediction: The absolute neutrino masses are closed-form D96 expressions with the normal ordering $m_1 = 0 < m_2 < m_3$ and $\Sigma m_\nu = 0.0581 \text{ eV} < 0.12 \text{ eV}$ (claim classification: correspondence + calibration).

Experimental Test: Direct mass measurement (KATRIN, next-generation tritium beta-decay) and cosmology (sum of masses from CMB/LSS).

Current Evidence: $\Sigma m_\nu = 0.0581 \text{ eV}$ is below current limits ($\Sigma m_\nu \lesssim 0.12 \text{ eV}$ from cosmology); the absolute scale is not yet measured; the mass ordering is not yet determined.

Future Validation Path: KATRIN sensitivity reach; JUNO mass-ordering determination; DESI/Euclid sum-of-masses bounds.

Falsification Criterion: A measured Σm_ν inconsistent with 0.0581 eV , or an inverted ordering established ($m_3 < m_2$).

AT-P010

Status: PENDING

Title: Neutrino mass ordering (normal).

Derived From: Same chain as AT-P009: the zero-mode mass $m_1 = 0$ fixes the normal ordering.

Prediction: The neutrino mass ordering is normal ($m_1 = 0 < m_2 < m_3$).

Experimental Test: Mass-ordering determination via long-baseline oscillations (DUNE, Hyper-K) and matter effects.

Current Evidence: Current data mildly prefer normal ordering; not conclusive.

Future Validation Path: DUNE (first half of the 2030s) and Hyper-K.

Falsification Criterion: Establishment of inverted ordering with significance.

AT-P011

Status: CONSISTENT

Title: Quark mass ratios (absolute masses).

Derived From: Difference $\rightarrow$ Actualization $\rightarrow$ Spectrum (D96) $\rightarrow$ unified mass law: $m_u = m_e \cdot \Sigma\sqrt{m}/\sqrt{\Sigma m^2} = 2.164$ (dev. 0.18%); $m_d = m_u \cdot (\Sigma\sqrt{m})^2/\text{occMom} = 4.676$ (dev. 0.14%); $m_s = m_d \cdot \text{occMom}/\Sigma m = 93.5$; $m_c = m_d \cdot (\Sigma\sqrt{m})^2/\sqrt{\Sigma m^2} = 1269$; $m_b = m_d \cdot \text{occMom}^2 \cdot \Sigma m \cdot \#g/(\Sigma\sqrt{m})^4 = 4186$; $m_t = m_u \cdot \text{occMom} \cdot \#d = 172704 \text{ MeV}$.

Prediction: All six absolute quark masses are reconstructed from the electron anchor and D96 spectral moments, each within 0.2% (claim classification: calibration via the anchor m_e).

Experimental Test: Measurement of the quark masses in the $\overline{\text{MS}}$ scheme.

Current Evidence: All six masses match the PDG $\overline{\text{MS}}$ values within 0.2%.

Future Validation Path: Lattice QCD refinements of the light-quark masses.

Falsification Criterion: A measured quark mass deviating from its spectral construction beyond 0.2%.

7 Majorana Character

AT-P012

Status: PENDING

Title: Neutrino Majorana character.

Derived From: Difference $\rightarrow$ Actualization $\rightarrow$ Spectrum (D96) $\rightarrow$ T3-only self-conjugate channel (48/95 modes), unique $Q = 0$ sector with no conserved charge, real mass matrix from the reflection automorphism.

Prediction: The neutrino is Majorana: self-conjugate in its T_3 -only channel, with no conserved charge and a real mass matrix (claim classification: correspondence).

Experimental Test: Observation of neutrinoless double beta decay ($0\nu\beta\beta$).

Current Evidence: No $0\nu\beta\beta$ observation; current limits $m_{\beta\beta} \lesssim 0.036\text{--}0.156$ eV are far above the predicted 2.02 meV.

Future Validation Path: nEXO and LEGEND-1000 ton-scale experiments.

Falsification Criterion: A conclusive Dirac-character determination (e.g. a fundamental conservation law excluding $0\nu\beta\beta$).

AT-P013

Status: PENDING

Title: Effective Majorana mass $m_{\beta\beta}$ (pre-registered P2).

Derived From: Difference $\rightarrow$ Actualization $\rightarrow$ Spectrum (D96) $\rightarrow$ PMNS reads (AT-P007) + absolute masses (AT-P009) + Majorana character (AT-P012): $m_{\beta\beta} = |m_1 c_{12}^2 c_{13}^2 + m_2 s_{12}^2 c_{13}^2 e^{i\alpha_2} + m_3 s_{13}^2 e^{-2i\delta_\nu}| = 2.02$ meV (computed 2.0222 meV), with $\alpha_2 = \alpha_3 = 0$.

Prediction: $m_{\beta\beta} = 2.02$ meV, dominated by the term $m_2 s_{12}^2 c_{13}^2 = 2.52$ meV and robust to the CP phase (pre-registered, QG191).

Experimental Test: Search for $0\nu\beta\beta$ at nEXO / LEGEND-1000 (half-life sensitivity $\sim 10^{28}$ yr).

Current Evidence: None — the prediction is below all current limits (0.036–0.156 eV).

Future Validation Path: nEXO / LEGEND-1000 ton-scale reach.

Falsification Criterion: A measured upper limit below 2.02 meV (significant exclusion) falsifies P2.

8 Electron g-2

AT-P014

Status: CONSISTENT

Title: Electron anomalous magnetic moment.

Derived From: Difference $\rightarrow$ Actualization $\rightarrow$ Spectrum (D96) $\rightarrow$ Schwinger term modulated by the lightest-octave fraction: $a_e = (\alpha/2\pi)(1 - (\text{occ}_0/\Sigma m)^2) = 1.159655 \times 10^{-3}$ (dev. 0.0003%).

Prediction: The electron anomaly is the Schwinger term with the negative spectral correction $1 - (\text{occ}_0/\Sigma m)^2$ (claim classification: correspondence).

Experimental Test: Precision measurement of a_e (Penning-trap experiments) and theoretical QED computation.

Current Evidence: Matches experiment within 0.0003%.

Future Validation Path: Improved QED tenth-order terms and fine-structure determination.

Falsification Criterion: A measured a_e inconsistent with the spectral value beyond the stated accuracy.

AT-P015**Status: CONSISTENT****Title:** Electron g-2 anomaly-free beyond Δa_e .**Derived From:** Same chain: $\Delta a_e = (\alpha/2\pi)^3 \text{span}^{1/4} (\text{occ}_0/\Sigma m)^3 = 1.86 \times 10^{-13} < 10^{-12}$.**Prediction:** No electron g-2 anomaly: the new-physics contribution is below 10^{-12} (claim classification: correspondence).**Experimental Test:** Comparison of measured a_e with the SM prediction at 10^{-12} precision.**Current Evidence:** No electron g-2 anomaly observed; the value matches SM QED.**Future Validation Path:** Continued improvement of α and the QED series.**Falsification Criterion:** An electron g-2 discrepancy exceeding $\Delta a_e = 1.86 \times 10^{-13}$.**AT-P016****Status: CONSISTENT****Title:** Muon g-2 (spectral D96 mechanism).**Derived From:** Difference $\rightarrow$ Actualization $\rightarrow$ Spectrum (D96) $\rightarrow$ same Schwinger-plus-spectral-fraction mechanism with the dense-bulk sign: $a_\mu = (\alpha/2\pi)(1 + \lambda_2/\Sigma m) = 1.16644 \times 10^{-3}$ (dev. 0.045%); $\Delta a_\mu = (\alpha/2\pi)^3 \text{span}^{1/4} = 2.494 \times 10^{-9}$ (dev. 0.24%).**Prediction:** The muon anomaly is the same D96 mechanism as the electron, with the positive dense-bulk correction $+\lambda_2/\Sigma m$ (opposite spectral end) (claim classification: correspondence).**Experimental Test:** Measurement of a_μ at Fermilab (Muon g-2) and J-PARC.**Current Evidence:** The central value matches within 0.045%; the anomaly magnitude is consistent with the measured g-2 tension band.**Future Validation Path:** Final Fermilab result; J-PARC.**Falsification Criterion:** A final measured a_μ inconsistent with the spectral value beyond the stated accuracy.

9 Oblique Parameters (S, T, U)

AT-P017**Status: CONSISTENT****Title:** Oblique parameters from spectral structure.**Derived From:** Difference $\rightarrow$ Actualization $\rightarrow$ Spectrum (D96) $\rightarrow$ lightest- octave fraction: $S = \text{occ}_0/\Sigma m = 4/95 = 0.0421$ (dev. 5.3%); $T = 2S = 8/95 = 0.0842$ (dev. 5.3%); $U = 0$ via the exact tree-level $\rho = 1$.**Prediction:** $S = 0.0421$, $T = 0.0842$ with $T = 2S$ exactly and $U = 0$ (claim classification: correspondence).**Experimental Test:** Electroweak global fit determination of S, T, U .**Current Evidence:** Consistent with the EW global fit beyond the masses and widths.**Future Validation Path:** Tightening of the EW fit at HL-LHC and future e^+e^- colliders.**Falsification Criterion:** An EW global fit excluding $S = 0.0421$, $T = 0.0842$, $U = 0$ at the stated accuracy.

Part III

Gravity Predictions

10 Gravitational Sector

AT-P018

Status: CONSISTENT

Title: Newton constant from the D96 spectral content.**Derived From:** Difference $\rightarrow$ Actualization $\rightarrow$ Spectrum (D96) $\rightarrow$ occupation- weighted content: $M_{\text{Pl}} = v \cdot (\Sigma m \cdot \#g \cdot \text{occ}_2)^3 = 1.22335 \times 10^{19} \text{ GeV}$ (dev. 0.20%); $G = 1/M_{\text{Pl}}^2 = 6.6476 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ (dev. 0.40%).**Prediction:** The Planck scale is the weak scale times the cube of the occupation-weighted spectral content; the physical exponent is cubic (QG183) (claim classification: calibration via the anchor v plus SI conversion).**Experimental Test:** Measurement of G (torsion balance) and M_{Pl} .**Current Evidence:** Matches G within 0.40%.**Future Validation Path:** Improved G determinations (CODATA).**Falsification Criterion:** A measured G inconsistent with the spectral construction beyond the stated accuracy.

AT-P019

Status: CONSISTENT

Title: Gravitational time dilation (GPS).**Derived From:** Difference $\rightarrow$ Actualization $\rightarrow$ counting measure $\rho \rightarrow$ redshift law: clock rate $d\tau/dt = \rho^{1/d} = \sqrt{-g_{00}}$; GPS offset $+38.5 \mu\text{s/day}$ vs. observed $+38.6 \mu\text{s/day}$ (dev. 0.2%).**Prediction:** Gravitational time dilation is the counting-measure redshift law; the GPS rate offset is -4.465×10^{-10} (claim classification: correspondence).**Experimental Test:** GPS satellite clock comparisons.**Current Evidence:** The net computed offset $+38.5 \mu\text{s/day}$ matches the observed $+38.6 \mu\text{s/day}$ within 0.2%.**Future Validation Path:** Continuing GPS and optical-clock tests.**Falsification Criterion:** A measured gravitational redshift inconsistent with the counting-measure law beyond the stated accuracy.

AT-P020

Status: CONSISTENT

Title: Frame dragging (Lense–Thirring).**Derived From:** Difference $\rightarrow$ tensor face ψ against $\eta \rightarrow$ restored gravitomagnetic sector: $\Omega_{\text{LT}} = G(3(\mathbf{J} \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{J})/(2c^2 r^3)$; GP-B 41.1 vs. 39.2 mas/yr; LAGEOS 30.7 vs. ~ 31 mas/yr.**Prediction:** Frame dragging is a ψ -sector observable at full GR strength when the tensor face is present (claim classification: correspondence).**Experimental Test:** GP-B and LAGEOS measurements of the Lense–Thirring precession.

Current Evidence: GP-B 41.1 vs. 39.2 mas/yr (+4.8%, inside the measured 37.2 ± 7.2); LAGEOS 30.7 vs. ~ 31 (−1.1%).

Future Validation Path: Future gravity-mission measurements (e.g. proposed LAGEOS-III-type and satellite tests).

Falsification Criterion: A measured frame-dragging precession inconsistent with the Lense–Thirring value at full GR strength.

11 Black-Hole Relations

AT-P021

Status: CONSISTENT

Title: Mass-radius relation $M \propto R$.

Derived From: Difference $\rightarrow$ Actualization $\rightarrow$ counting measure $\rightarrow$ per-octave (logarithmic) deficit profile: $a \propto -1/r$, $GM_{\text{eff}} \propto R$ (claim classification: correspondence; the flat-rotation-curve profile is a hosted input).

Prediction: The observed $M \propto R$ relation follows from the counting measure: the deficit is per-octave/log, giving the flat rotation-curve profile and $GM_{\text{eff}} \propto R$.

Experimental Test: Galactic rotation curves and supermassive black-hole mass–radius correlations.

Current Evidence: Flat rotation curves are observed; $M \propto R$ is consistent with the hosted profile.

Future Validation Path: High-precision rotation curves (Rubin LSST) and horizon-scale mass measurements (EHT).

Falsification Criterion: A measured rotation-curve profile inconsistent with the per-octave deficit profile across scales.

AT-P022

Status: CONSISTENT

Title: Hawking temperature $T \propto 1/R$.

Derived From: Same chain: with $S \propto R^{d-1}$ and $M \propto R$, the first law gives $T \propto 1/R$; the ψ sector contributes only a radius-independent prefactor.

Prediction: The Hawking temperature scales as $T \propto 1/R$; the $T(R_1)/T(R_2)$ ratio is ψ -invariant (QG208).

Experimental Test: Event-horizon-scale observations; astrophysical black-hole cooling bounds.

Current Evidence: Consistent with the standard $T = \hbar\kappa/(2\pi k_B)$ structure and current bounds.

Future Validation Path: Gravitational-wave ringdown precision and horizon-scale measurements.

Falsification Criterion: A measured black-hole temperature inconsistent with $T \propto 1/R$ beyond the stated accuracy.

12 Bekenstein Boundary

AT-P023
Status: CONSISTENT (documented boundary)

Title: Bekenstein structure $S \propto A$; exact $1/4$ coefficient is a boundary.

Derived From: Difference $\rightarrow$ Actualization $\rightarrow$ counting measure: $S \propto A$ (QG12), $M \propto R$ (AT-P021), $T \propto 1/R$ (AT-P022) fully derived; the exact coefficient $S = A/4$ is *not* derivable from D96/TRM — it requires the imported quantum factor $T = \kappa/(2\pi)$ (QG185, QG196 impossibility proof).

Prediction: The *structure* of the Bekenstein law ($S \propto A$, $M \propto R$, $T \propto 1/R$) is a derived consequence; the exact $1/4$ coefficient is a disclosed boundary, not a derivation claim.

Experimental Test: Tests of the area law and the entropy coefficient (astrophysical and analogue systems).

Current Evidence: The structure is consistent with black-hole thermodynamics; the coefficient is not tested at the required precision.

Future Validation Path: Quantum-gravity-sensitive observations; the coefficient remains a documented boundary.

Falsification Criterion: This entry is a boundary statement, not a falsifiable prediction; it documents what the theory does not derive.

Part IV

Cosmology Predictions

13 Ω_Λ

AT-P024
Status: CONSISTENT

Title: Vacuum density fraction.

Derived From: Difference $\rightarrow$ Actualization $\rightarrow$ Spectrum (D96) $\rightarrow$ octave occupancies [4, 4, 87]: $\Omega_\Lambda = I_{\text{occ}}/\ln K_{\text{oct}} = 0.6839$ (dev. 0.12%); $I_{\text{occ}} = KL(p||\text{uniform}) = 0.7513$ nats; $K_{\text{oct}} = 3$ occupied octave classes.

Prediction: The vacuum fraction is the information-density fraction of the D96 octave record (claim classification: correspondence).

Experimental Test: Cosmological probes of the dark-energy fraction (CMB, supernovae, BAO).

Current Evidence: $\Omega_\Lambda = 0.6839$ matches the observed value within 0.12%.

Future Validation Path: DESI, Euclid, and Rubin LSST precision cosmology.

Falsification Criterion: A measured Ω_Λ inconsistent with 0.6839 beyond the stated accuracy.

14 Ω_m

AT-P025

Status: CONSISTENT

Title: Matter density fraction.

Derived From: Same chain: $\Omega_m = 1 - \Omega_\Lambda = 0.3161$ (dev. 0.26%).

Prediction: The matter fraction is the complement of the octave-record information fraction (claim classification: correspondence).

Experimental Test: Cosmological probes of the matter density (CMB, BAO, galaxy clustering, lensing).

Current Evidence: $\Omega_m = 0.3161$ matches the observed value within 0.26%.

Future Validation Path: DESI, Euclid, and Rubin LSST precision cosmology.

Falsification Criterion: A measured Ω_m inconsistent with 0.3161 beyond the stated accuracy.

15 Acoustic Peaks

AT-P026

Status: CONSISTENT

Title: First acoustic peak existence (structural).

Derived From: Difference $\rightarrow$ Actualization $\rightarrow$ Spectrum (D96) $\rightarrow$ recombination- scale mode ladder: the fundamental doublet at $\omega_{\min} = 0.6216$ is isolated by the dominant spectral gap (ln-gap 0.680, largest of 94).

Prediction: A first peak exists as a structural consequence: the fundamental doublet is the maximally isolated low-frequency spectral feature and therefore the unique D96 candidate for a first peak (claim classification: theorem — structural).

Experimental Test: Observation of the CMB angular power spectrum.

Current Evidence: The CMB first peak is observed at $\ell_1 \simeq 220$. Existence is structural; the identification with the observed peak and its multipole position require the ℓ -axis normalization.

Future Validation Path: CMB-S4 and Planck-precision acoustic spectra.

Falsification Criterion: Absence of a first peak in the CMB angular spectrum.

AT-P027

Status: CONSISTENT

Title: First peak location ℓ_1 (fitted normalization).

Derived From: Same chain: $\ell_1 = \Sigma m \cdot \ln(\text{span}) \cdot (5/4) = 220.48$ (dev. 0.008%).

Prediction: $\ell_1 = 220.48$; the multiplier 5/4 is a documented *fit* for the ℓ -axis normalization (QG297), not part of the selection mechanism (claim classification: fit).

Experimental Test: Measurement of the first-peak multipole.

Current Evidence: Observed $\ell_1 \simeq 220.48$; the value matches because the 5/4 factor is fitted to it.

Future Validation Path: None beyond precision cosmology; the location is a documented fit, not a derivation.

Falsification Criterion: Not applicable — the location is a documented fit, not a falsifiable derivation.

AT-P028
Status: CONSISTENT

Title: Peak ratios (spectral correspondences).

Derived From: Same chain: $r_{21} = (\Sigma m - \#d) \cdot \text{occ}_0 / \text{occ}_2 = 2.4368$ (dev. 0.035%), $r_{31} = \text{span} / \sqrt{3} = 3.6965$ (dev. 0.058%).

Prediction: The ratios of the second and third peak positions to the first are pure spectral ratios, with no fitted constant (claim classification: correspondence).

Experimental Test: Measurement of ℓ_2/ℓ_1 and ℓ_3/ℓ_1 .

Current Evidence: Matches observation within $\sim 0.05\%$.

Future Validation Path: CMB-S4 precision.

Falsification Criterion: A measured peak ratio inconsistent with 2.4368 or 3.6965 beyond the stated accuracy.

AT-P029
Status: CONSISTENT

Title: Scalar spectral index n_s and running.

Derived From: Same chain: $n_s = 0.96497$ (dev. 0.007%), running = 0 (QG237); seed spectrum is the Poisson counting variance, scale-free from critical branching.

Prediction: $n_s = 0.96497$ with zero running; the seed is the counting variance (claim classification: correspondence).

Experimental Test: Measurement of the scalar spectral index and its running from the CMB power spectrum.

Current Evidence: Planck measures $n_s = 0.9649 \pm 0.0042$; running consistent with zero.

Future Validation Path: CMB-S4.

Falsification Criterion: A measured running significantly different from zero, or an n_s inconsistent with 0.96497 beyond the stated accuracy.

16 Structure Formation

AT-P030
Status: CONSISTENT

Title: Cosmological constant existence, sign, and scaling.

Derived From: Difference $\rightarrow$ Actualization $\rightarrow$ critical branching vacuum $\rightarrow$ residual actualization pressure: at criticality ($\mu = 1$) the Galton–Watson mean is constant while the variance grows, so the realized vacuum never equals its uniform expectation.

Prediction: Λ exists, is positive, and scales with the spectral/actualization structure (claim classification: correspondence).

Experimental Test: Measurement of the cosmological constant and its equation of state.

Current Evidence: $\Lambda > 0$ observed; $w \simeq -1$ consistent.

Future Validation Path: DESI/Euclid dark-energy characterization.

Falsification Criterion: A measured $\Lambda \leq 0$, or a dark-energy dynamics inconsistent with a constant residual pressure.

AT-P031

Status: PENDING

Title: Structure formation from Q-event statistics.

Derived From: Difference $\rightarrow$ Actualization $\rightarrow$ Q-event statistics and spectral hierarchy $\rightarrow$ large-scale organization.

Prediction: The large-scale organization of the universe follows the actualization dynamics: structure formation is derived from the Q-event statistics and the spectral hierarchy, with the initial conditions the uniform critical state and the information content the departure that seeds the structure (claim classification: correspondence).

Experimental Test: Galaxy-clustering statistics, the matter power spectrum, and the CMB lensing signal.

Current Evidence: Qualitative consistency with the standard structure-formation picture; no discriminating quantitative test yet performed.

Future Validation Path: DESI, Euclid, and Rubin LSST matter-power-spectrum measurements.

Falsification Criterion: A large-scale structure signal inconsistent with the Q-event-statistics prediction at discriminating precision.

AT-P032

Status: CONSISTENT

Title: Scale-free seed spectrum (Poisson counting variance).

Derived From: Same chain: $\delta_i = 1/\sqrt{\langle N \rangle}$, scale-free ($n_s = 1$ at the seed level) from critical branching.

Prediction: The seed power spectrum is the Poisson counting variance, scale-free from critical branching (claim classification: correspondence).

Experimental Test: Primordial power-spectrum shape from the CMB.

Current Evidence: Consistent with the near-scale-invariant observed spectrum ($n_s = 0.96497$).

Future Validation Path: CMB-S4 precision.

Falsification Criterion: A primordial spectrum strongly inconsistent with the counting-variance seed.

Part V

Universality Predictions

17 Operator Basis

AT-P033

Status: CONSISTENT

Title: Four-operator basis universal across organized domains.

Derived From: Difference $\rightarrow$ Actualization $\rightarrow$ Spectrum (D96) $\rightarrow$ operator layer: Crowding (degeneracy), Compression (octaves), Beat (extent), Locking (gaps) as spectrum projections.

Prediction: The four-operator basis recurs as the structural read of organized spectra in every organized domain (claim classification: correspondence).

Experimental Test: Quantitative operator reads of organized systems: networks, language, music, DNA, software, finance.

Current Evidence: The four operators appear across the seven tested domains (QG313); random spectra fail the basis while organized systems carry it (QG309, QG312).

Future Validation Path: New organized-domain datasets.

Falsification Criterion: An organized domain whose spectrum fails all four operators at the quantitative level.

AT-P034

Status: CONSISTENT

Title: No fifth operator.

Derived From: Same chain: fifth-operator search (QG307) across unexplored domains.

Prediction: No genuine fifth spectral operator exists beyond Crowding, Compression, Beat, Locking with the universal MOMENT read-out (search-scoped, not an existence proof).

Experimental Test: Search for an additional operator in organized-domain spectra.

Current Evidence: No genuine fifth operator found in any searched domain; candidates reduce to compositions or are not spectral reads.

Future Validation Path: Continued domain expansion.

Falsification Criterion: A genuine fifth operator (not a composition) in an organized-domain spectrum.

18 Lock Law

AT-P035

Status: CONSISTENT

Title: Lock law: universal structure, domain-specific values.

Derived From: Difference $\rightarrow$ Actualization $\rightarrow$ Spectrum (D96) $\rightarrow$ Locking operator $\rightarrow$ lock identities (normalized moment ratios).

Prediction: Every organized domain carries stable, reproducible lock identities (universal structure); the lock values are domain-specific fingerprints (claim classification: correspondence).

Experimental Test: Measurement of the moment/span, compression/count, higher-moment, and $\sqrt{\text{moment}}/\text{span}$ ratios across organized domains.

Current Evidence: Stable lock structure in all seven tested domains (physics, language, music, DNA, software, finance, networks); the D96 locks $\text{occMom}/\Sigma m = 20.0026$, $\Sigma m^2/\Sigma m = 2.4105$, $\text{occMom}/\Sigma m^2 = 8.2980$ follow the moment-chain identity exactly.

Future Validation Path: New organized-domain cohorts.

Falsification Criterion: An organized domain with no stable lock identity at the quantitative level.

AT-P036**Status: CONSISTENT****Title:** Locks precede organization.**Derived From:** Same chain: early-lock prediction (QG315) on evolving systems.**Prediction:** In evolving systems, the lock structure appears before the full organization develops, and systems with early lock structure reorganize less under structural change (cohort-scoped on synthetic deterministic evolving-law cohorts).**Experimental Test:** Evolutionary cohorts (software history, wiki edits, citation networks, language corpora).**Current Evidence:** Lock identity reaches half-strength earlier than maturity in most systems; locked groups are 100% small-reorganization.**Future Validation Path:** Real-world evolutionary datasets.**Falsification Criterion:** An evolving system whose lock structure lags the maturity development.

19 Organization Transition

AT-P037**Status: CONSISTENT****Title:** Organization as a phase transition.**Derived From:** Same chain: organization-phase-transition audit (QG316) sweeping a continuous organization parameter across the four regimes.**Prediction:** The binary operator basis completes discontinuously at the critical parameter $g^* \simeq 0.31$, while the quantitative organization grows continuously; the locks appear at the critical onset (cohort-scoped).**Experimental Test:** Controlled organization sweeps (white noise $\rightarrow$ strong organization).**Current Evidence:** The all-four operator screen flips at $g^* \simeq 0.31$ and persists for all stronger organization; the lock coherence emerges in the same window.**Future Validation Path:** Real-world organization transitions.**Falsification Criterion:** A continuous organization transition with no critical threshold for the binary operator basis.

AT-P038**Status: CONSISTENT****Title:** Blind prediction of future organization class.**Derived From:** Same chain: blind-organization prediction (QG314) — organization score from lock structure only, rule fixed before the reveal.**Prediction:** The future organization class can be predicted from the early lock structure with the prediction rule fixed in advance; the lock values predict the class, not the strength within it.**Experimental Test:** Blind classification of future high-maturity systems from early-stage lock coherence.

Current Evidence: The early lock presence identifies the future top-class systems exactly; the score separates the organized from the unorganized class.

Future Validation Path: Prospective organizational datasets.

Falsification Criterion: A future-class prediction failure beyond the stated accuracy of the locked protocol.

Part VI

Experimental Frontier

20 Predictions Awaiting Validation

The three pre-registered predictions P1, P2, and P3 are the experimental frontier of the theory. They are frozen in the immutable registry (QG193); their values never change and only their outcome states advance.

AT-P039

Status: PENDING

Title: P1 — 106 GeV resonance.

Derived From: Difference $\rightarrow$ Actualization $\rightarrow$ Spectrum (D96) $\rightarrow$ sector ladder $\rightarrow$ missing-Z-anchor rung: $M_{106} = 7 \cdot M_Z / 6 = 7 \cdot 15.198 \text{ GeV}$; window $= M_{106} \pm (\text{rung spacing})/2 = 99\text{--}114 \text{ GeV}$ (stated 98.79–113.99).

Prediction: A resonance at 106.39 GeV central mass with window 99–114 GeV; production hierarchy 9 rungs $106.4 \rightarrow 263.4 \text{ GeV}$; decay hierarchy unit quantum $15.20 \text{ GeV} \times 10 + \text{top } 20.26 \text{ GeV} \times 1$ (pre-registered, QG190; forbidden-input guard: no ATLAS/CMS/fitted-mass constant).

Uncertainty: $\pm 7.60 \text{ GeV}$ (half the mean rung spacing); boson-anchor family agrees within 0.74%.

Experimental Test: LHC Run 3 and HL-LHC diphoton/diboson searches in the 99–114 GeV window.

Current Evidence: PENDING: the window is neither confirmed nor excluded. The $\sim 95 \text{ GeV}$ scalar excess cluster aligns with the 91.19 GeV rung (dev. 4.5%), not with the 106.39 GeV prediction; CMS/ATLAS full-Run-2 diphoton null searches cover 106 GeV with limits 15–102 fb; LEP2 excludes SM-like $< 114.4 \text{ GeV}$ but SM-strength hZZ only.

Future Validation Path: HL-LHC 3000 fb^{-1} (projected 1–3 fb sensitivity in 100–106 GeV); decisive.

Falsification Criterion: No signal in statistically sensitive searches of the 99–114 GeV window (DISFAVORED/FALSIFIED).

AT-P040

Status: PENDING

Title: P2 — $0\nu\beta\beta$ effective Majorana mass.

Derived From: Difference $\rightarrow$ Actualization $\rightarrow$ Spectrum (D96) $\rightarrow$ PMNS reads (AT-P007) + absolute masses (AT-P009) + Majorana character (AT-P012): $m_{\beta\beta} = |m_1 c_{12}^2 c_{13}^2 + m_2 s_{12}^2 c_{13}^2 e^{i\alpha_2} +$

$m_3 s_{13}^2 e^{-2i\delta_\nu} = 2.02 \text{ meV}$ (computed 2.0222 meV), $\alpha_2 = \alpha_3 = 0$ (pre-registered, QG191; forbidden-input guard: no experimental limit or detector sensitivity).

Uncertainty: $\pm 10\%$ (1.8–2.2 meV); dominated by $m_2 s_{12}^2 c_{13}^2 = 2.52 \text{ meV}$, robust to the CP phase.

Experimental Test: $0\nu\beta\beta$ search at nEXO / LEGEND-1000 (half-life sensitivity $\sim 10^{28} \text{ yr}$).

Current Evidence: PENDING: below all current limits (0.036–0.156 eV).

Future Validation Path: nEXO / LEGEND-1000 ton-scale reach.

Falsification Criterion: A measured upper limit below 2.02 meV (significant exclusion) falsifies P2.

AT-P041

Status: PENDING

Title: P3 — Sector-ladder spectrum.

Derived From: Difference $\rightarrow$ Actualization $\rightarrow$ Spectrum (D96) $\rightarrow$ ladder radii $\times$ Z-anchor scale: $E_{\text{rung}} = \text{radius} \cdot (M_Z/6)$; unit quantum $\Delta E = M_Z/6 = 15.20 \text{ GeV}$, top quantum 20.26 GeV (pre-registered, QG192; forbidden-input guard: no collider bump or catalog).

Prediction: Nine resonances: $106.39 \text{ (primary)} \rightarrow 136.78 \rightarrow 151.98 \rightarrow 182.38 \rightarrow 197.58 \rightarrow 212.78 \rightarrow 227.97 \rightarrow 243.17 \rightarrow 263.43 \text{ GeV}$, multiplicities unit $\times 10 + \text{top} \times 1$, width scale 15.20 GeV.

Uncertainty: $\pm 5\%$ per rung; boson-anchor family agrees within 0.74%.

Experimental Test: High-energy searches at LHC Run 3 / HL-LHC and future FCC-hh.

Current Evidence: PENDING overall; the 151.98 GeV rung matches the combined $\sim 152 \text{ GeV}$ diphoton excess (local 3.6σ , global up to 5.4σ ; alignment 0.0132% deviation, look-elsewhere-corrected $z = 2.80\sigma$); SM anchors Z/H/t confirm the ladder scale (QG200); eight rungs remain unprobed.

Future Validation Path: HL-LHC confirmation of the 152 GeV excess and full Run-3 searches of the remaining rungs.

Falsification Criterion: A sensitive search excludes any frozen rung (limit below the rung energy) falsifies P3.

Registry rule: no future phase may modify a registered prediction (QG193). Only the outcome state may advance: PENDING $\rightarrow$ SUPPORTED $\rightarrow$ CONFIRMED, or PENDING $\rightarrow$ DISFAVORED $\rightarrow$ FALSIFIED. Frozen values never change.

Status Summary

ID	Status	Prediction	Key value
AT-P001	CONSISTENT	Gauge structure 1 + 3 + 8	$12 = \deg C_{96}$
AT-P002	CONSISTENT	Gauge couplings	137; 3/95; 8/64.08; 0.2316
AT-P003	CONSISTENT	Weak boson masses	M_W 80.1, M_Z 91.4 GeV
AT-P004	CONSISTENT	Precision EW	$\sin^2 \theta_{\text{eff}} = 0.23158$
AT-P005	CONSISTENT	CKM	dev. 0.58%
AT-P006	CONSISTENT	CKM CP	$\delta_{CP} = 66.3^\circ$, $J = 3.14 \times 10^{-5}$
AT-P007	CONSISTENT	PMNS	dev. 1.5%
AT-P008	CONSISTENT	Neutrino splittings	7.607×10^{-5} ; 2.44×10^{-3} eV ²
AT-P009	PENDING	Absolute neutrino masses	8.72; 49.4 meV; $\Sigma m_\nu = 0.0581$ eV
AT-P010	PENDING	Mass ordering (normal)	$m_1 = 0$
AT-P011	CONSISTENT	Quark masses	all six within 0.2%
AT-P012	PENDING	Majorana character	self-conjugate channel
AT-P013	PENDING	$m_{\beta\beta}$ (P2)	2.02 meV
AT-P014	CONSISTENT	Electron g-2	1.159655×10^{-3}
AT-P015	CONSISTENT	Electron anomaly-free	$\Delta a_e = 1.86 \times 10^{-13}$
AT-P016	CONSISTENT	Muon g-2	1.16644×10^{-3}
AT-P017	CONSISTENT	Oblique parameters	$S = 0.0421$, $T = 0.0842$, $U = 0$
AT-P018	CONSISTENT	Newton constant	6.6476×10^{-11} m ³ /kg/s ²
AT-P019	CONSISTENT	GPS time dilation	+38.5 μ s/day
AT-P020	CONSISTENT	Frame dragging	GP-B 41.1 mas/yr
AT-P021	CONSISTENT	Mass-radius $M \propto R$	$GM_{\text{eff}} \propto R$
AT-P022	CONSISTENT	Hawking $T \propto 1/R$	ψ -invariant ratio
AT-P023	BOUNDARY	Bekenstein structure	1/4 coefficient not derived
AT-P024	CONSISTENT	Ω_Λ	0.6839
AT-P025	CONSISTENT	Ω_m	0.3161
AT-P026	CONSISTENT	First peak existence	structural
AT-P027	CONSISTENT	First peak location	$\ell_1 = 220.48$ (fit)
AT-P028	CONSISTENT	Peak ratios	2.4368; 3.6965
AT-P029	CONSISTENT	Spectral index	$n_s = 0.96497$, running = 0
AT-P030	CONSISTENT	Λ existence/sign	$\Lambda > 0$
AT-P031	PENDING	Structure formation	Q-event statistics
AT-P032	CONSISTENT	Seed spectrum	scale-free counting variance
AT-P033	CONSISTENT	Four-operator universality	seven domains
AT-P034	CONSISTENT	No fifth operator	search-scoped
AT-P035	CONSISTENT	Lock law	universal structure, domain values

Registry totals			
CONSISTENT	PENDING	BOUNDARY	FALSIFIED
32	8	1	0

No registered prediction is currently falsified. The three pre-registered frontier predictions (AT-P039–AT-P041) are the experimental targets.

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